

Contoso Electronics

Equipment Reference & Maintenance Guide

Pick and Place Machine

Contoso RapidPlace PP-3200

Asset Identification

Field	Detail
Asset ID	CE-PNP-2200
Category	SMT Line
Manufacturer	Contoso Assembly Tools
Location	SMT Line 1 - Position 3
Installation Date	2021-11-30
Warranty Expiry	2026-11-30
Support Contact	smt.line.support@contoso.com ext. 4501

Overview

The Contoso RapidPlace PP-3200 is a high-speed SMT pick-and-place machine that mounts surface-mount components onto printed circuit boards with vision-guided precision. It supports a wide component range from 01005 chips to large connectors and BGAs.

Technical Specifications

Parameter	Value
Placement Speed	45,000 CPH (optimal)
Placement Accuracy	±25 µm @ 3σ
Component Range	01005 to 55 x 55 mm
Feeder Capacity	120 8mm feeders
Vision System	Dual on-head + fixed fiducial camera
PCB Size	50 x 50 to 510 x 460 mm
Power	3-phase 400 VAC, 6 kW

Preventive Maintenance Schedule

Interval	Task
Daily	Clean nozzles and check vacuum; verify feeder seating.
Weekly	Lubricate XY gantry; clean camera lenses.
Monthly	Run placement accuracy calibration; inspect nozzle wear.

Quarterly	Replace worn nozzles and feeder springs; check belts.
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Safety Guidelines

- Keep hands clear of the gantry - moving parts can cause injury.
- Never bypass light curtains or door interlocks.
- Follow ESD procedures when handling components and boards.
- Lockout/tagout before reaching into the machine envelope.

Common Issues & Troubleshooting

Symptom	Recommended Action
Components not picked	Check nozzle vacuum, feeder pitch, and part height in library.
Placement offset	Recalibrate vision fiducials; verify PCB seating.
Frequent tombstoning	Review paste deposit and placement force; check nozzle.
Feeder errors	Reseat feeder; clean feeder contacts; replace if faulty.

Contoso Electronics internal maintenance documentation. For service requests, contact smt.line.support@contoso.com | ext. 4501 or submit a work order through the maintenance portal.